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The Land Speed Record honours the vehicle that, simply put, moves the fastest on plain ground. Thrust SSC, a British jet-powered car, set the first supersonic land speed record in 1997. It screamed its way across the Nevada desert at 759.333 mph on its first run, and at 766.109 mph on the second attempt. And that's on land—we are not talking about a plane here. Compare this with the speed of a typical Boeing 747 in cruise mode—about 500 mph.

However, those of us who live in the digital world are likely to be more impressed by the following: "Caltech and CERN top new performance threshold by sending 859 GB at more than 6.6 Gbps—in less than 17 minutes—across nearly 16,000 km." That's the new Internet2 (I2) Land Speed Record, and that speed is about 10,000 times faster than a typical north American home broadband connection.

Faster, Further

859 GB across 16,000 kilometres in 17 minutes. Internet2 is the beginning of an Internet on hyper-steroids

Networks are capable of much more than what the Internet is today—a congested place—there is no infrastructure for critical applications to take precedence over spam, for example. Neither is there the infrastructure for cutting-edge applications such as high-performance, low-latency videoconferencing.

Today's Internet will grow and evolve, and could eventually become something like I2.

What is I2?

I2 is a collaborative effort at developing advanced networking technology and applications important to research and education. Over 140 US universities, working in collaboration with partners in industry and the US government, are leading the I2 project.

The project does not aim at replacing the Internet: it is working to enable advanced applications such as digital libraries, telemedicine, virtual laboratories, etc., that are not possible with today's technologies.

I2 is not a separate network; it joins member network applications and engineering development efforts, together with many advanced campus, regional, and national networks. I2 is currently US-based, and US-centric. In fact, it is touted as something to help America retain world Internet dominance. However, the project is open to collaboration from international organisations, and there is no reason why we won't see I2-like networks everywhere, in a few years. The bandwidth is there; the technologies exist; it's only a matter of putting them to use.

What's the big difference?

The first (and most obvious) difference with I2 is raw speed. The Abilene backbone, one of the two that supports I2, "is a proving ground for high-bandwidth technologies. The cross-country backbone is 10 gigabits per second, with the goal of offering 100 megabits per second of connectivity between every Abilene connected desktop," according to internet2.edu. See box "Abilene, Mean, Stronger" for more information.

Another change is that I2 is a collaborative, research-and-education effort; it is not currently used for commercial purposes. Also, I2 is an experimental testbed for several technologies and techniques. There are no bandwidth restrictions—in fact, members and affiliates pay a flat fee, and are encouraged to use the bandwidth available.

So why, in the first place, do we need something more than today's Internet? There are several answers.

God Speed

Networks are capable of much faster speeds than we generally imagine. Broadcasts over I2 can go at 250 megabytes per second. IPv6 (See box "IPv6, And What It Means For The Future") will support such speed and applications. For example, IPv6 supports between 64 KB and 4 GB. (The current version of the